



NARAYANA
IIT ACADEMY
INDIA

41
YEARS
OF EXCELLENCE

JEE ADVANCED GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

45
YEARS
OF EXCELLENCE

Sec: OSR.IIT_*CO-SC
Time: 3HRS

Date: 07-04-24
Max. Marks: 180

Name of the Student: _____

H.T. NO:

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07-04-24_OSR.STAR CO-SUPER CHAINA_JEE-ADV_GTA-4(P2)_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 6)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 7 – 12)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 13 – 16)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	-1	4	12
Sec – IV (Q.N : 17 – 19)	Questions with Non-Negative Integer type	+4	0	3	12
Total				19	60

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 20 – 25)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 26 – 31)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 32 – 35)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	-1	4	12
Sec – IV (Q.N : 36 – 38)	Questions with Non-Negative Integer type	+4	0	3	12
Total				19	60

CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 39 – 44)	Questions with Multiple Correct Choice (partial marking scheme) (+1,0)	+4	-2	6	24
Sec – II (Q.N : 45 – 50)	Questions with Comprehension Type With Numerical value type (3 Comprehensions – 2 + 2 + 2 = 6Q)	+2	0	6	12
Sec – III (Q.N : 51 – 54)	Questions with Comprehension Type (2 Comprehensions – 2 + 2 = 4Q)	+3	-1	4	12
Sec – IV (Q.N : 55 – 57)	Questions with Non-Negative Integer type	+4	0	3	12
Total				19	60

MATHEMATICS**MAX.MARKS: 60****SECTION - 1 (Maximum Marks : 24)**

This section contains **SIX (06)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

1. $I_n = \int_{\frac{n}{2}}^{\frac{n+1}{2}} \frac{\sin(\pi \sin^2 \pi x)}{(\sqrt{2})^x} dx, n \in \mathbb{Z}$ then

A) $\frac{I_n}{I_{n+4}} = 2$

B) $\frac{I_n}{I_{n+4}} = \frac{1}{\sqrt{2}}$

C) $\frac{\sum_{n=0}^{\infty} I_{8n}}{I_0} = \frac{4}{3}$

D) $\frac{\sum_{n=0}^{\infty} I_n}{I_0} = 2$

2. A parallelopiped is formed using 3-non coplanar vectors $\vec{a}, \vec{b}, \vec{c}$ with fixed magnitudes angles between any of the vector with normal of the plane determined by the other two is

α and the volume of parallelopiped is T and its surface area is y. If $\left(\frac{y}{T}\right) = 4 \left(\frac{1}{|\vec{a}|} + \frac{1}{|\vec{b}|} + \frac{1}{|\vec{c}|}\right)$

then

A) $\cos^2 \alpha + \cos \alpha = \frac{3}{4}$

B) $\sin^2 \alpha + \sin^4 \alpha = \frac{21}{16}$

C) $\cos^2 \alpha + \cos \alpha = \frac{3+2\sqrt{3}}{4}$

D) $\sin^2 \alpha + \sin^4 \alpha = \frac{5}{16}$

3. Z_1, Z_2, Z_3 are 3 non-zero distinct points satisfying $|Z-1|=1$ and $Z_2^2 = Z_1 Z_3$ then

A) $\frac{Z_3 - Z_2}{Z_2 + Z_3 - 2}$ is purely imaginary

B) $\arg\left(\frac{Z_2 - 1}{Z_3 - 1}\right) = 2 \arg\left(\frac{Z_2}{Z_3}\right)$

C) $\arg\left(\frac{Z_2 - 1}{Z_1 - 1}\right) = 2 \arg\left(\frac{Z_3}{Z_1}\right)$

D) $\left|\frac{1}{Z_2} - \frac{1}{Z_3}\right| + \left|\frac{1}{Z_1} - \frac{1}{Z_2}\right| = \left|\frac{1}{Z_1} - \frac{1}{Z_3}\right|$

4. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be two functions $f(x) = |x| \lfloor x^2 - 1 \rfloor$, $g(x) = |x| + \lfloor x^2 - 1 \rfloor$. Where $|\cdot|$ is modulus function and $\lfloor \cdot \rfloor = \text{G.I.F.}$, then which of the following options are true ?

A) f is discontinuous exactly at 5 points of $[-1, 2]$

B) g is discontinuous exactly at 5 points of $[-1, 2]$

C) $\int_{-\frac{1}{2}}^{\frac{1}{2}} (f(x) - g(x)) dx = \frac{1}{2}$

D) $\int_{-\frac{1}{2}}^{\frac{1}{2}} (f(x) - g(x)) dx = 1$

5. The lengths of two perpendicular tangents from a point to a parabola are 3 and 4 then length of latus rectum of parabola is

A) $\frac{125}{144}$

B) $\frac{144}{125}$

C) $\frac{576}{125}$

D) $\frac{125}{576}$

6. The first term of an infinite G.P. is 21. The 2nd term and sum of the series are both positive integers. All possible values of 2nd term are

A) 12

B) 14

C) 18

D) 20

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +2 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 7 and 8:

$$A(Z_0) = ai, B(Z_1) = \frac{1}{2} + bi, C(Z_2) = 1 + ci, a, b, c \in \mathbb{R}.$$

W_1 represents Z when A, B, C are non collinear.

W_2 represents Z when A, B, C are collinear where

$$Z = Z_0 \cos^4 t + 2 Z_1 \cos^2 t \cdot \sin^2 t + Z_2 \sin^4 t \quad (t \in \mathbb{R}).$$

7. If $b < a$ ($a, b \in \mathbb{R}^+$) then number of points of intersection of W_2 with $Z = t + ibt^2$ ($t \in \mathbb{R}$) is ____

8. If $b > a$ ($a, b \in \mathbb{R}^+$) then number of points of intersection of W_2 with the curve $Z = t + ibt^2$ ($t \in \mathbb{R}$) is _____

Question Stem for Question Nos. 9 and 10:

Let $f: R \rightarrow R$ be a function defined as $f(x) = 3x^5 - 25x^3 + 60x + 5$ and

$$g(x) = \begin{cases} \max\{f(t) : -4 \leq t \leq x\} & ; \quad -4 \leq x \leq 0 \\ \min\{f(t) : 0 < t \leq x\} & ; \quad 0 < x \leq 2 \\ f(x) - 16 & ; \quad x > 2 \end{cases}$$

9. Total number of points at which $y = f(x)$ and $y = g(x)$ is non differentiable in $[-4, \infty)$ is _
10. If $f(x) + \lambda = 0$ has exactly 3 distinct real roots then number of integral values of $\lambda =$ _____

Question Stem for Question Nos. 11 and 12:

A person repeatedly tosses a fair dice. He gets two points for a throw of a perfect square and one point otherwise. Let P_n denote the probability that he reaches a score of exactly n then

11. $\frac{P_{10} - P_9}{P_8 - P_9} =$ _____

12. $\lim_{n \rightarrow \infty} P_n =$ _____

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.

Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Negative Marks: -1 In all other cases.

Question Stem for Question Nos. 13 and 14:

Let $X = \{1, 2, 3\}$. A number is selected from the set X with replacement and this process is repeated 1100 times for each $i, i \in \{1, 2, 3\}$. Let $f(i)$ represent the number of times the number i is selected. Also let S denote the total sum of 1100 numbers selected. If $S^3 = 162 f(1) \cdot f(2) \cdot f(3)$ then

13. Harmonic mean of $f(1), f(2), f(3)$ is _____

A) 100

B) 200

C) 300

D) 400

14. There are $f(1)$ red balls, $f(2)$ blue balls, $f(3)$ white balls, X green balls. (balls of same colour are identical). The balls are arranged in a row such that no two balls of same colour are consecutive. Let X_1 and X_2 both minimum and maximum values of X respectively for which above arrangement is possible then $X_1 + X_2 =$ _____
- A) 1000 B) 1100 C) 1200 D) 1300

Question Stem for Question Nos. 15 and 16:

Let $y = f(x)$ be a solution of $\frac{dy}{dx} = -4x(y-2)$ with $f(0) = 2$ and

$$g(x) = \max \left\{ 1 + \frac{x}{2} - \left\lfloor \frac{x}{2} \right\rfloor, \sin x, |x-1| \right\}, h(x) = \log_e x \quad ([.] \rightarrow G.I.F.)$$

15. Area enclosed by $y = f(x)$, $y = g(x)$ and y-axis.
- A) 1 B) 2 C) 3 D) 4
16. Area bounded by the curve $y = f(x)$, $y = h(x)$ and the y-axis is
- A) e B) e^2 C) $\frac{3e}{2}$ D) $e^2 - 1$

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**. For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

17. If $(2\sin^2 x - 2\sqrt{3}\sin x \cos x + 1)(9 - 4\cos^2 x - 4\cos x) = 40$. The number of values of x in $[0, 20]$ is _____
18. Let F be the foot of perpendicular from $P(1, 2, -3)$ on the line $L: \frac{x+1}{2} = \frac{y-3}{-2} = \frac{z}{-1}$. Given $Q(x_1, y_1, z_1)$ and $R(x_2, y_2, z_2)$ are points at a distance 3 units from F on the line L. Then $\frac{1}{24} \left(\sum_{i=1}^2 x_i^2 + y_i^2 + (z_i)^2 \right) =$ _____
19. Tangents are drawn from any point on the straight line $y = x + 8$ to the auxiliary circle of $x^2 + 16y^2 = 16$. If A and B are points of contact of these tangents and P, Q are corresponding points of A and B on the ellipse respectively. If locus of mid point of PQ is $x^2 + \alpha y^2 + \beta x + \gamma y = 0$ then $\alpha - \beta + \gamma =$ _____

PHYSICS**MAX.MARKS: 60****SECTION - 1 (Maximum Marks : 24)**

This section contains **SIX (06)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

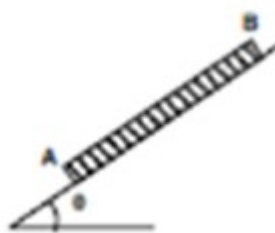
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

20. A uniform metal rod (AB) of mass m and length L is lying on a rough incline. The inclination of the incline is θ and coefficient of friction between the rod and the incline is μ . Choose the correct options.



- A) If temperature increases the rod expands. However, there is a point P on the rod which does not move. The distance of this point from the lower end of the rod is

$$\frac{L}{2} \left[1 + \frac{\tan \theta}{\mu} \right].$$

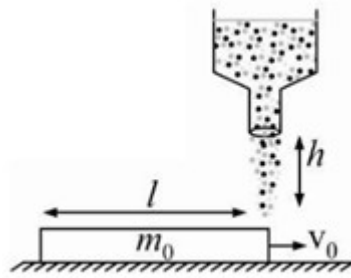
- B) If the temperature falls the rod contracts. Once again there is a point Q which does

not move. The distance of Q from the lower end of the rod is $\frac{L}{2} \left[1 - \frac{\tan \theta}{\mu} \right]$

- C) Repeated expansion and contraction cause the rod to slide down

- D) Repeated expansion and contraction will not cause the rod to slide down

21. A plank of mass m_0 and length ℓ moving with velocity v_0 on a smooth horizontal surface passes under a stationary hopper. Sand spills from the hopper at a constant rate of $\mu \text{ kg/s}$, falls through height h onto the plank and sticks to it. Let v be the instantaneous velocity of plank, choose the correct option(s).



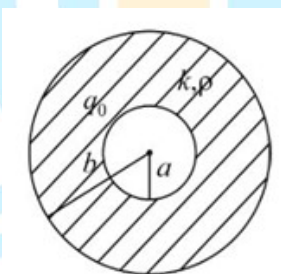
A) Horizontal component of force exerted by falling sand on the plank is proportional to v .

B) Vertical component of force exerted by falling sand on the plank is proportional to $\sqrt{h}$

C) Acceleration of the plank is proportional to v

D) Velocity of the plank after it completely passes from under the hopper is $v_0 e^{\frac{-\mu l}{m_0 v_0}}$.

22. The radii of a spherical capacitor are equal to a and b ($b > a$). The space between the two spherical shells is filled with a dielectric of dielectric constant K and resistivity ρ . At $t = 0$, the inner sphere is given a charge q_0 . Choose the correct option(s).



A) Charge q on the inner sphere as a function of time is given by $q = q_0 e^{-\frac{t}{\rho K \epsilon_0}}$

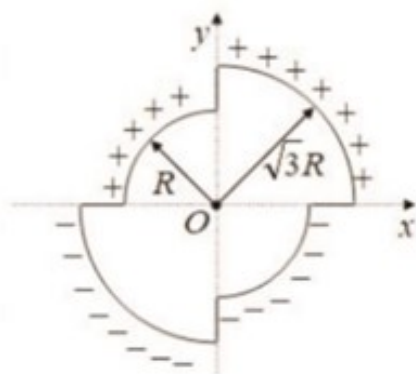
B) The charge on the inner sphere will become zero almost instantly

C) After a long time, the charge on the outer sphere will become q_0

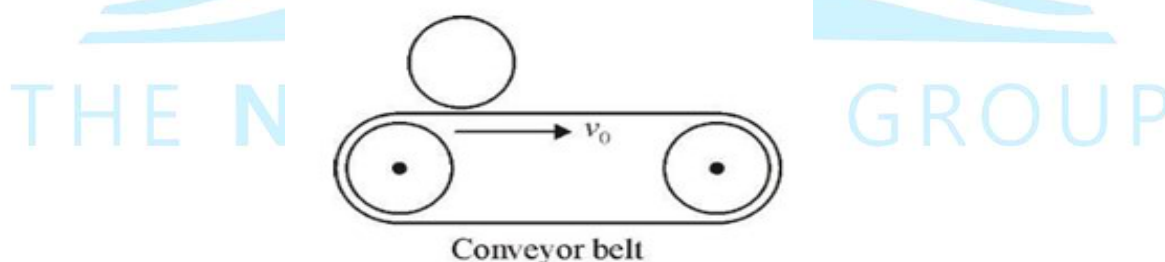
D) The total amount of heat generated during the redistribution of charge will be given

by $H = \left(\frac{1}{a} - \frac{1}{b} \right) \frac{q_0^2}{8\pi\epsilon_0}$

23. In the figure there are four arcs carrying positive or negative charge all of them have same magnitude of charge density λ . Choose the correct statement(s).



- A) The net dipole moment for the given charge distribution is $4\sqrt{5}\lambda R^2$
- B) The resultant electric field at the center and at any point on the axis passing through point O and perpendicular to the plane of arcs are opposite in direction
- C) If a uniform electric field is switched on perpendicular to the plane then the charge distribution starts rotating about x-axis
- D) Potential at the center of the given charge distribution is zero
24. A stationary homogenous sphere of radius R and mass M is gently placed on the conveyor belt moving with a constant velocity v_0 towards the east. After a time t from the instant the sphere was placed on the belt, it starts pure rolling and its CM moves with a velocity v relative to the ground. Coefficient of kinetic friction is μ_k .



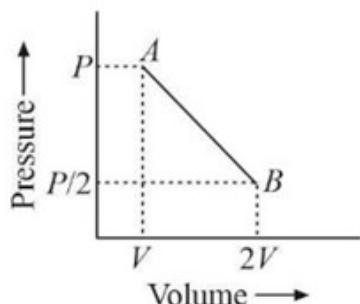
A) $t = \frac{2v_0}{5\mu_k g}$

B) $t = \frac{2v_0}{7\mu_k g}$

C) $v = \frac{2v_0}{5}$

D) $v = \frac{2v_0}{7}$

25. An ideal monoatomic gas is taken from state A (pressure P , volume V) to state B (pressure $P/2$, volume $2V$) along a straight line in the $P-V$ diagram as shown in figure. Then, choose the correct options.



- A) The work done by the gas in the process A to B exceeds the work that would be done by it if the system were taken from A to B along the isotherm
- B) In going from $A \rightarrow B$ the volume of gas where nature of process changes from endothermic to exothermic is $\frac{15}{8}V$
- C) In the $P-T$ diagram, the path AB becomes a part of a parabola
- D) In going from A to B, the temperature T of the gas first decreases to a minimum and then increases

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

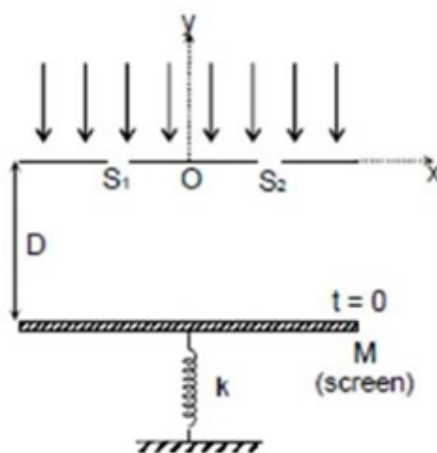
Full Marks: +2 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 26 and 27:

A monochromatic light of wavelength λ is incident on two slits S_1 and S_2 made on x -axis. The mid point of the line S_1S_2 is considered as origin and 'd' is the distance between the two slits which are symmetric about y -axis. The interference pattern is observed on a plate of mass M which is acting as a screen. The screen is attached to one end of a vertical spring of spring constant 'k' and the other end of spring is attached with the ground. Spring is massless and screen always remains in horizontal plane. The distance between the plane of slits and the screen is $D(D \gg d)$. Assuming S_1 and S_2 are coherent in nature. At $t = 0$, screen is released and spring is in its natural length. It is

given that $M = 1 \text{ kg}$, $k = 1 \text{ N/m}$, $D = \frac{5Mg}{k}$, $g = 10 \text{ m/s}^2$, $d = 200 \lambda$. The x co-ordinate (in cm) of the first order maxima on the screen as a function of time is given by $\pm(\lambda_1 - \lambda_2 \cos t)$.



26. The value of λ_1 is _____.

27. The value of λ_2 is _____.

Question Stem for Question Nos. 28 and 29:

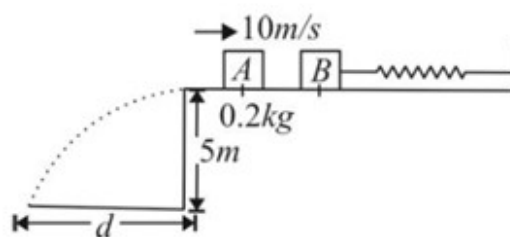
A galvanometer is used as an ammeter and a voltmeter. The range of the voltmeter can be changed to n times its original value with the help of a 27Ω multiplier. The range of the same galvanometer when used as an ammeter can be changed to n times its original value using a 3Ω shunt. Power dissipated by the galvanometer when giving a full-scale reading is $9 \times 10^{-4} \text{ W}$.

28. The resistance of the coil (in Ω) is _____.

29. The full scale deflection current (in mA) is _____.

Question Stem for Question Nos. 30 and 31:

Figure shows block A of mass 0.2 kg sliding to the right over a frictionless elevated surface at a speed of 10 m/s . The block undergoes a collision with stationary block B, which is connected to a non deformed spring of spring constant 1000 Nm^{-1} . The coefficient of restitution between the blocks is 0.5 . After the collision, block B oscillates in SHM with a period of 0.2 . Block A eventually slides off the opposite end of the elevated surface, landing a distance ' d ' from the base of that surface after falling height 5 m . (use $\pi^2 = 10$; $g = 10 \text{ m/s}^2$)



30. Amplitude of the SHM as being executed by block B spring system is $2.5\sqrt{x}$ cm, where x is ____.
31. The distance 'd' (in m) will be equal to ____.

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.

Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

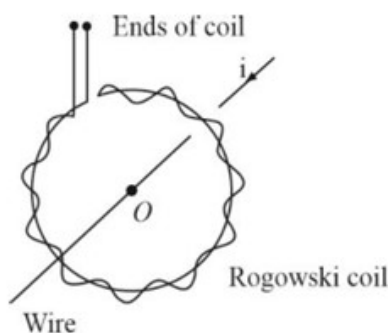
Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Negative Marks: -1 In all other cases.

Question Stem for Question Nos. 32 and 33:

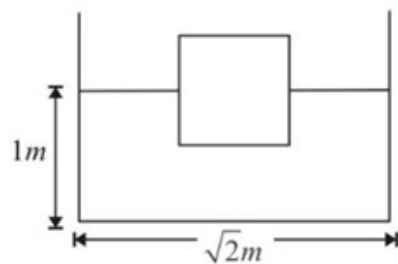
Rogowski coil: The coil shown in the figure, called Rogowski coil, is used in electrical measurements. The coil consists of a toroidal conductor wrapped around a circular return cord. The coil can be used to measure amplitude of alternating current or value of direct current in a wire. The coil is simply clipped around the wire. Measured value of emf across the ends of the coil gives value of current amplitude, in case a.c. is flowing through the wire. To measure direct current, ends of the coil are connected to ballistic galvanometer. Current in the wire is switched off and amount of charge crossing the loop is measured by the ballistic galvanometer. This value of charge is used to calculate value of direct current in the wire.



32. Number of turns per unit length of the coil, area of each loop in the coil, resistance of the coil, are n , S and R respectively. Current flowing through the wire crossing the plane of the coil is i . A ballistic galvanometer is connected across the ends of the coil and the current is switched off, charge that flows through the coil is measured to be Q . The correct relation is
- A) $\frac{\mu_0 n S i}{QR} = 1$ B) $\frac{\mu_0 n S i}{QR} = 2$ C) $\frac{\mu_0 n S i}{QR} = \frac{1}{2}$ D) $\frac{\mu_0 n S i}{Q^2 R} = 1$
33. Consider two current carrying wires carrying currents i_1 and i_2 in opposite directions. Both the wires are perpendicular to the plane of the coil. Wire carrying currents i_1 and i_2 are respectively at a distance $0.5r$ and $1.5r$ from the centre of the coil. Reading of ballistic galvanometer depends on $[r : \text{mean radius of the coil}]$
- A) i_1 B) $i_1 - i_2$ C) $\frac{i_1}{2}$ D) $\frac{i_1 - i_2}{2}$

Question Stem for Question Nos. 34 and 35:

A cube of wood of side 1m and density 500 kg/m^3 is dipped in a tank of water (density 1000 kg/m^3). We want to push the cube to the bottom of the tank. The depth of the water in tank is 1m , as shown figure. The cross section of the base of tank is $\sqrt{2}\text{m} \times \sqrt{2}\text{m}$ and the water does not spill out as the cube is pushed down.



34. As the cube is pushed down in the tank:
- A) The force required to push it down increases continuously.
- B) The force required to push it down decreases continuously.
- C) The force required to push it down increases and then becomes constant.
- D) The force required to push it down decreases and then becomes constant.

35. The minimum work required to be done by the person pushing the cube to the base of the tank is

A) 1250 J

B) 2500 J

C) 1875 J

D) 625 J

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

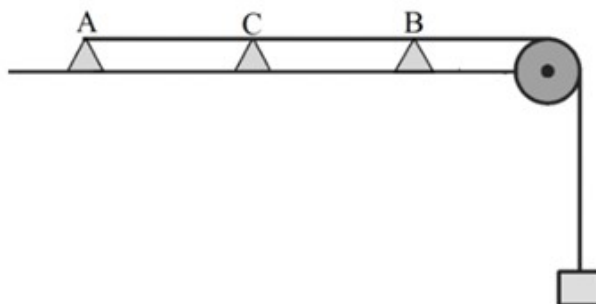
Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

36. Two radioactive samples A_1 and A_2 having half life 3 years and 2 years respectively have been decaying for many years. Today the number of atoms in the sample A_1 is twice the number of atoms in the sample A_2 . Both the samples had same number of atoms X years ago, calculate X.

37. A thin plano-convex lens acts like a concave mirror of radius of curvature 20cm when its plane surface is silvered. The radius of curvature of the curved surface (in cm) if index of refraction of its material is 1.5 will be 'R', then $\frac{R}{2} = \underline{\hspace{2cm}}$

38. A thin metallic wire of mass per unit length 0.02kg/m is clamped at three points A, B and C and a block of weight 50 N is suspended from the wire as shown. The length $AC = 25$ cm and the length $CB = 26$ cm. If both segments (AC and CB) of the wire are made to vibrate in their fundamental modes, the number of beats heard per second (to the nearest integer) is .



CHEMISTRY**MAX.MARKS: 60****SECTION - 1 (Maximum Marks : 24)**

This section contains **SIX (06)** questions.

Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).

For each question, choose the correct option(s) to answer the question.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks: -2 In all other cases.

39. Which of the following is (are) correct.

A) $[NiCl_4]^{-2}$ is paramagnetic while $[Ni(CO)_4]$ is diamagnetic though both are tetrahedral.

B) The colour of $[Ti(H_2O)_6]^{+3}$ is pale yellow.

C) $[Fe(H_2O)_6]^{+3}$ is strongly paramagnetic whereas $[Fe(CN)_6]^{-3}$ is weakly paramagnetic.

D) $[Co(NH_3)_6]^{+3}$ is an inner orbital complex whereas $[Ni(NH_3)_6]^{+2}$ is an outer orbital complex.

40. Which of the following transition metals are having more melting point than 'Tc'

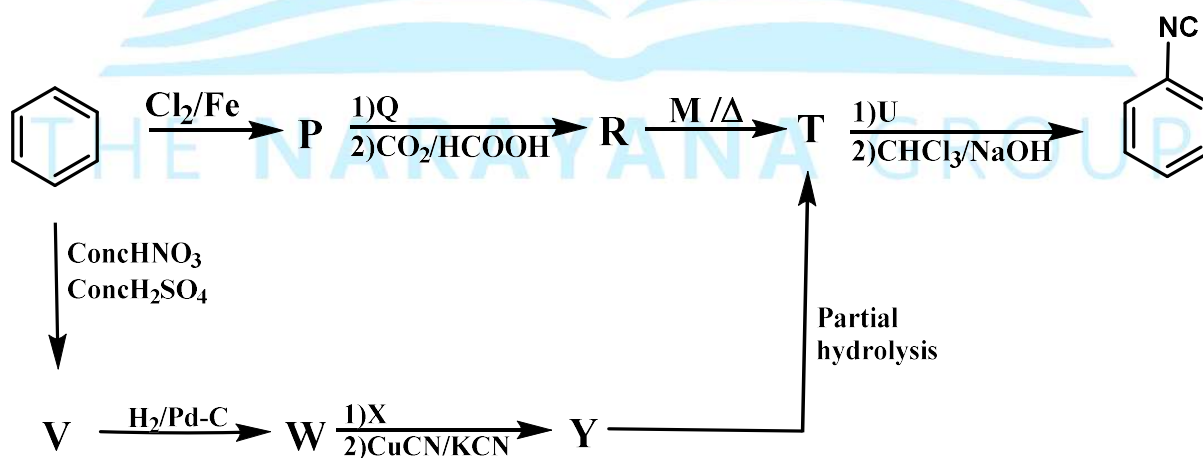
A) Rh

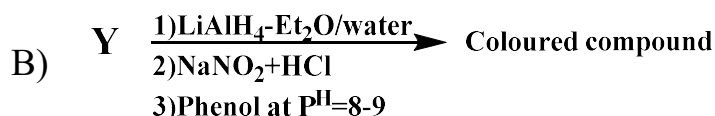
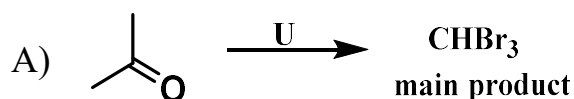
B) V

C) Mo

D) Re

41. Choose the correct statement(s) regarding given reaction scheme.





42. Which of the following combinations would result in the formation of a buffer solution.

- A) $\text{NH}_3 + \text{NH}_4\text{Cl}$
B) $\text{CH}_3\text{COOH} + \text{NaOH}$ in 2 : 1 molar ratio
C) $\text{NH}_3 + \text{HCl}$ in 2 : 1 molar ratio
D) $\text{CH}_3\text{COOH} + \text{NaOH}$ in 1 : 2 molar ratio

43. The correct order of electron affinity of the elements/ions is/are

- A) $\text{O} > \text{Se}$ B) $\text{F}^+ > \text{Cl}^+$ C) $\text{P} > \text{N}$ D) $\text{Ne} > \text{Ar}$

44. What is true about ice ?

- A) Its density is more than water
B) It is good conductor of heat
C) It is electrical insulator (poor conductor)
D) $\Delta_f H_{\text{ice}}^0 = 0$

SECTION - 2 (Maximum Marks : 12)

This section contains **TWO (3)** Paragraphs. Based on each paragraph, there are **2** questions.

The answer to each question is a **NUMERICAL VALUE**.

For each question, enter the correct numerical value of the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer. If the numerical value has more than two decimal places **truncate/round-off** the value to **TWO** decimal places.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +2 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

Question Stem for Question Nos. 45 and 46:

Phosphorous when treated with excess of chlorine forms compound 'X'. This 'X' when reacts with phosphorous pentoxide form a product 'Y'.

45. The number of sp^3 hybridized atoms present in one molecule of the product formed when 'Y' is dissolved in excess of hot water.
46. In the following reaction oxidation state of 'Z' is

**Question Stem for Question Nos. 47 and 48:**

As a result of the isobaric heating by $\Delta T = 72K$, one mole of a certain ideal gas obtains an amount of heat $Q = 1.598 kJ$.

47. The |work| performed by the gas is _____ J.
48. The increment of its internal energy is _____ kJ.

Question Stem for Question Nos. 49 and 50:

A volume of 100 ml of 0.1 M H_3PO_4 solution is titrated with 0.5M $NaOH$ solution till the second equivalence point. Then 10 ml of 0.5 M HCl solution is mixed in the resulting solution. The dissociation constants of H_3PO_4 are 10^{-3} , 10^{-8} and 10^{-13} .

49. pH at the second equivalence point is _____
50. pH of the solution after adding HCl is _____

SECTION - 3 (Maximum Marks : 12)

This section contains **TWO (2)** Paragraphs. Based on each paragraph, there are **2** questions.

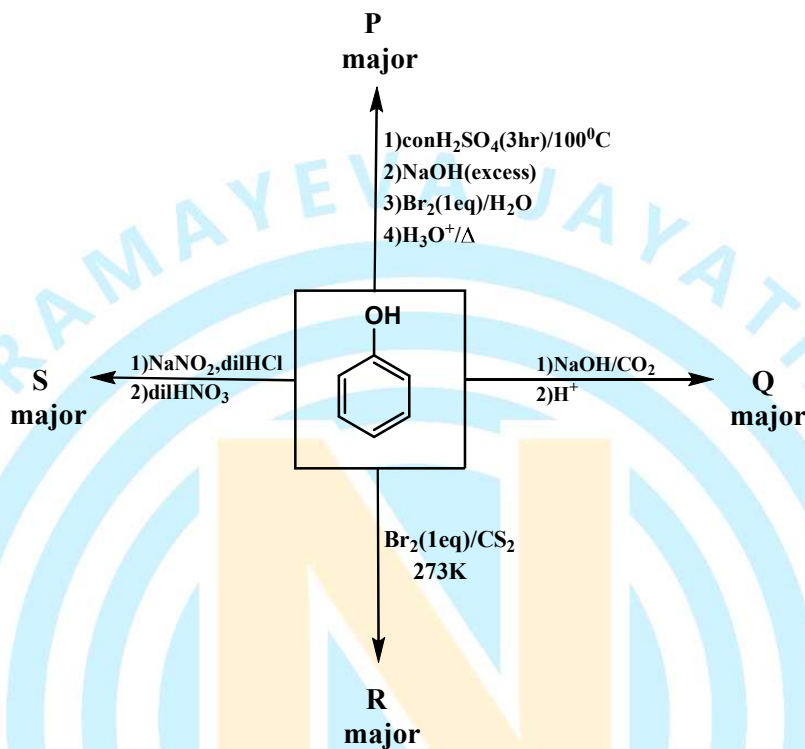
Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer. For each question, choose the option corresponding to the correct answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +3 If **ONLY** the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

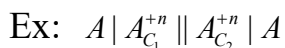
Negative Marks: -1 In all other cases.

Question Stem for Question Nos. 51 and 52:

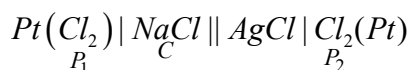
51. P and R are
 A) Identicals B) Chain isomers C) Positional isomers D) Functional isomers
52. Which of the following is more acidic.
 A) P B) R C) Q D) S

Question Stem for Question Nos. 53 and 54:

Concentration cell is one in which anode and cathode of a voltaic cell is made of same electrode.



$C_1 \neq C_2$ to have some emf in the cell. Now consider the cell.



$K_{sp} \text{ AgCl is } 10^{-12}$

53. If $P_1 = P_2$, required condition so that the above cell has positive emf is

- A) $C > 10^{-6} M$ B) $C < 10^{-6} M$ C) $P_1 = P_2 = 1 \text{ atm}$

D) Emf of the cell cannot be found as it is not a concentration cell

54. If $C = 10^{-6}$ which of the following conditions will give positive emf.

- A) $P_1 > P_2$ B) $P_1 < P_2$ C) $P_1 = P_2$ D) None

SECTION - 4 (Maximum Marks : 12)

This section contains **THREE (03)** questions. The answer to each question is a **NON-NEGATIVE INTEGER**

For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter answer.

Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4 If ONLY the correct numerical value is entered as answer.

Zero Marks: 0 In all other cases.

55. The value of (Y - X) in the following reaction.



56. 'X' is the number of essential amino acids of the following.

Glycine, Leucine, Glutamic acid, Aspartic acid, Lysine, Serine, Cysteine, Methionine, Tyrosine, Phenyl alanine, Proline.

'Y' is the number of D-sugars of the following structural formula.



Then $(X+Y) / 2$ is

57. The total number of unpaired electrons present in N_2^+ , O_2^{+2} , N_2^{-2} are _____